

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

Apply morphological rules to strip affixes and identify the base form.

Handle both grammatical variations and word formation changes.

Ensure consistent root extraction across all forms.

Determine the stem and classification for each input word.

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.

Apply state transitions to represent suffix transformations and irregular forms in verb inflection.

Differentiate between regular and irregular inflectional patterns within the dataset.

Ensure consistent mapping of all forms to a common root representation.

Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.

Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.

Handle variations in suffix patterns while preserving semantic consistency.

Ensure uniform root extraction across all derived forms.

Determine the stemming steps, intermediate forms, and final stem for each input word.

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system. Apply rules to decompose each word into root and suffix components. Differentiate between inflectional and derivational forms within the dataset. Normalize all variants to a common base form for efficient retrieval. Determine the parsed structure and normalized output for each word.

Q1. Morphological Analysis of "connected", "connecting", "connection"

Step 1: Rule-Based Decomposition

Word	Root	Suffix	Type
Connected	connect	-ed	Inflectional
Connecting	connect	-ing	Inflectional
Connection	connect	-ion	Derivational

Step 2: Classification

- **-ed** → Past tense marker (Inflectional)
- **-ing** → Present participle (Inflectional)
- **-ion** → Forms a noun from a verb (Derivational)

Step 3: Normalization

All three words are normalized to the common base form:

connect

Parsed Structure

Connected

└─ Root : connect
└─ Suffix : -ed

Connecting

└─ Root : connect
└─ Suffix : -ing

Connection

└─ Root : connect
└─ Suffix : -ion

Final Output

Word	Root	Suffix	Classification	Normalized Form
Connected	connect	-ed	Inflectional	connect
Connecting	connect	-ing	Inflectional	connect
Connection	connect	-ion	Derivational	connect

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis. Identify prefixes, suffixes, and base forms using rule-based decomposition. Ensure semantic consistency across derived word forms. Classify each transformation as inflectional or derivational. Produce the root and morphological breakdown for each word.

Q2. Morphological Parsing of "unhappy", "happiness", "happily"

Step 1: Rule-Based Decomposition

Word	Prefix	Root	Suffix
unhappy	un-	happy	—
happiness	—	happy	-ness
happily	—	happy	-ly

Step 2: Semantic Analysis

- **un-** changes meaning to negative.
- **-ness** converts adjective into noun.
- **-ly** converts adjective into adverb.

Step 3: Classification

Word	Type
unhappy	Derivational
happiness	Derivational
happily	Derivational

Morphological Breakdown

unhappy

└ Prefix : un-
└ Root : happy

happiness

└ Root : happy
└ Suffix : -ness

happily

└ Root : happy
└ Suffix : -ly

Final Output

Word	Prefix	Root	Suffix	Classification
unhappy	un-	happy	—	Derivational
happiness	—	happy	-ness	Derivational
happily	—	happy	-ly	Derivational

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine. Apply morphological rules to strip affixes and identify the base form. Handle both grammatical variations and word formation changes. Ensure consistent root extraction across all forms. Determine the stem and classification for each input word.

Q3. Stemming of "played", "player", "playing"

Step 1: Morphological Rules

Word	Rule Applied
played	Remove -ed
player	Remove -er
playing	Remove -ing

Step 2: Stem Extraction

played

↓ Remove -ed

play

player

↓ Remove -er

play

playing

↓ Remove -ing

play

Step 3: Classification

Word	Type
played	Inflectional
player	Derivational
playing	Inflectional

Final Output

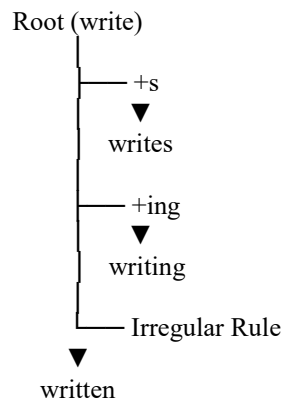
Word	Affix Removed	Stem	Classification
played	-ed	play	Inflectional
player	-er	play	Derivational
playing	-ing	play	Inflectional

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system. Apply state transitions to represent suffix transformations and irregular forms in verb inflection. Differentiate between regular and irregular inflectional patterns within the dataset. Ensure consistent mapping of all forms to a common root representation. Determine the state transitions, morphological breakdown, and normalized output for each word.

Q4. Finite-State Morphological Parsing of "writes", "writing", "written"

Finite-State Transitions





Morphological Breakdown

Word	Root	Suffix	Pattern
writes	write	-s	Regular Inflection
writing	write	-ing	Regular Inflection
written	write	-en	Irregular Inflection

Normalization

All words normalize to

write

Final Output

Word	State Transition	Classification	Normalized Form
writes	write → +s	Regular Inflection	write
writing	write → +ing	Regular Inflection	write
written	write → irregular (-en)	Irregular Inflection	write

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval. Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms. Handle variations in suffix patterns while preserving semantic consistency. Ensure uniform root extraction across all derived forms. Determine the stemming steps, intermediate forms, and final stem for each input word.

Q5. Porter Stemmer Analysis of "relational", "relation", "relate"

Step 1: Porter Stemming Rules

Word 1: relational

relational

↓ Remove -ational

relate

↓ Remove -e

Relat

Word 2: relation

relation

↓ Remove -ion

relat

Word 3: relate

relate

↓ Remove final -e

Relat

Intermediate Forms

Word	Intermediate Form	Final Stem
relational	relate	relat
relation	relat	relat
relate	relat	relat

Classification

All three are derivational because suffixes create different word forms without changing the core meaning.

Final Output

Word	Stemming Rule	Intermediate Form	Final Stem
relational	Remove -ational	relate	relat
relation	Remove -ion	relat	relat
relate	Remove final -e	relat	relat

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____